

Resolving Coordination Frictions in Green Labor Transitions

Minimizing Unemployment, Fiscal Costs, and Welfare Distortions

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Agenda

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Motivation I: Green Transition is about people

- **Green transition:** Economy-wide shift from fossil fuels to low-carbon to achieve net-zero target by 2050, [?]
- It is not only about technology and capital; it is about people.
- Demands worker reallocation at scale and speed.
⇒ Costly frictions and bottlenecks
- Political economy of green transition often cites:
 - ▶ Unemployment consequences
 - ▶ Funding costs of policies

Motivation I: Unemployment concerns

Unemployment concerns amidst the fossil-fuel workers



Biden Faces More Pressure From Environmentalists to Block Steel Merger

Climate change joins national security and concern about jobs in a mounting pressure campaign to prevent Nippon Steel from buying U.S. Steel.

The New York Times

The Energy Transition Is Underway. Fossil Fuel Workers Could Be Left Behind.

The Biden administration is trying to increase renewable energy investments in distressed regions, but some are skeptical those measures would be enough to make up for job losses.

The New York Times

Labour's net zero quest will cost jobs, unions fear

11 September 2024

BBC

CLIMATE CHANGE

Fossil-Fuel Industry Workers Say Green Economy Could Leave Them in Dust

Many environmental and labor leaders pitch New York's clean energy transition as a way to employ young and marginalized people – little consolation to workers in fossil-fuel industries worried about job loss.

Financial costs concerns of green policies

Future of UN climate dialogue threatened by **budget shortfall**

By Kate Abnett

October 25, 2024 2:05 PM EDT • Updated October 25, 2024



SEPTEMBER 5, 2023 6:47AM

The Inflation Reduction Act's Energy Subsidies Are More **Expensive** Than You Think

Around the world, backlash against **expensive** climate change policies

By H. Sterling Burnett

May 18, 2019 4:00 am

ENERGY, ENVIRONMENT, FEDERAL GOVERNMENT

Coalition urges Congress to end **costly** green subsidies, citing economic harm

Who can **afford** to go green? Hard-pressed consumers are pushing back



Analysis by [Lizanna Zandy](#), CNN

9 minute read • Published 6:17 AM EST, Tue November 20, 2023

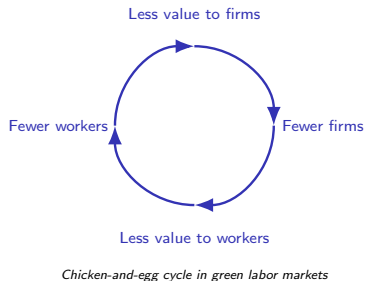
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Article | [Open access](#) | Published: 30 June 2021

Higher cost of finance exacerbates a climate investment trap in developing economies

Motivation II: Green transition still requires coordination

- **Even once you decide to transition, core issue:** a *two-sided* coordination problem.
- **Mechanism:** workers pay entry cost & face job uncertainty; firms earn subsidy only post-match and delay vacancies
⇒ each waits ⇒ under-entry/under-vacancy.
- **Evidence:** postings outpace green talent; hiring bottlenecks; displacement risk; 2030 shortfalls [Bluedorn et al., 2023, ?, ?] [LinkedIn Economic Graph](#).
- **Motivation:** policies should be designed to ease the underlying coordination issue.



What do I do

- **Question:** How to best design policy mix that most efficiently expands green employment share from **2% → 14%** in the US by 2030?
- **Policy Instruments:** Entry subsidy to workers (s_g); production subsidy to green firms (τ_g); combination of both; balanced budget via tax.
- **Compare them based on:**
 - ▶ Unemployment consequences
 - ▶ Fiscal cost (% of GDP)
 - ▶ Aggregate Welfare
- **Approach:**
 - ▶ Two-sector Diamond-Moretessen-Pissarides model
 - ▶ Calibration: U.S. labor moments in 2022.
 - ▶ Study steady-state transition from 2% → 14%
 - ▶ Study dynamics along the path $L_g \in \{2, \dots, 14\}\%$.

What we find

- **Insight 1:** Coordinated policy dominates one-sided to meet the target:
 - ▶ Unemployment: **-39 to -2 bps**
 - ▶ Fiscal cost: **-5 to -3 bps of GDP**
 - ▶ Welfare (CEV): **+23 to +27 bps**

Mechanism: Dual-subsidy eases the underlying coordination issue.

- **Insight 2:** Policy mix changes along the path:
 - ▶ Unemployment and fiscal support are hump-shaped
 - ▶ Mix rotates: worker → firm → worker (early → mid → late)

Mechanism: Binding margin changes with market tightness.

- **Insight 3:** Aggregate welfare improves only above a threshold
 - ▶ Transition raises aggregate welfare if the environmental benefit $\geq 0.87\%$ (CEV)

Mechanism: Green transition needs to balance productivity-externality trade-off.

- Policy takeaways:
 - ▶ Use both instruments to be socially inclusive and fiscally sustainable
 - ▶ Adjust the mix with tightness and the binding margin
 - ▶ Evaluate net gains against carefully calculated externality threshold

Literature & gaps

Labor, reallocation, and the green transition

- **CGE: ??????**
- **CGE + labor frictions:** ??Hafstead and Williams III [2018], Hafstead et al. [2022]
- **Search & matching: ???**
- **Environmental macro: ?????**
- **Two-sided markets: ?**
- **Reports: ????**

Gaps I address

- *Focus on the two-sided coordination issue*
- *Effective Policy design focus*
- *Environmental externalities in DMP model*

Static: misalignment and policy margin

- Extend the standard one-shot Diamond Mortensen Pissarides (DMP) model
- Two sectors: **green** (g) and **non-green** (n)²; firms open vacancies, workers search.
- Firms pay a vacancy cost c and workers pay an entry cost κ_g .
- Matches produce an amount p of the numeraire good; unemployed receive benefit $z < p$.
- Wages are set by Nash bargaining (worker bargaining power η).
- **Friction:** Misalignment between who bears the cost and benefits is set up
 - ▶ Workers entering green requires an up-front cost κ_g
 - ▶ Firms producing in green subject to production subsidy τ_g
- **Policy instruments:**
 - ▶ Entrant grant s_g (reduces effective κ_g)
 - ▶ Production subsidy τ_g to green firms (paid only when a match produces)
- Government runs a *balanced budget*: finances τ_g with a tax.

²Following Curtis and Marinescu [2023], we define green jobs as renewable-energy occupations (solar, wind, EVs) for clear calibration and policy relevance; these roles account for roughly 80% of U.S. energy-related emissions [?].

Static Model: Set-up

- *Workers*: Need to pay for entry cost to enter the green sector

Green: $\underbrace{-\kappa_g}_{\text{worker's entry cost}} + \alpha_{wg}w_g + (1 - \alpha_{wg})z$ and Brown: $\alpha_{wn}w_n + (1 - \alpha_{wn})z$.

- *Firms*: Gets the green production subsidy in the green sector

Green: $-c + \alpha_{fg}(y \underbrace{+\tau_g}_{\text{production subsidy to firms}} - w_g)$ and Brown: $-c + \alpha_{fn}(y - \tau_n - w_n)$.

- *Government*: Funds the subsidy τ_g to all matched and producing green firms by imposing a tax τ_n on all matched and producing brown firms:

$$\tau_n = \frac{\tau_g \alpha_{fg} v_g}{\alpha_{fn} v_n} = \tau_g \cdot \frac{\frac{\pi}{(\pi + v_g)} \cdot v_g}{\frac{(1 - \pi)}{(1 - \pi + v_n)} \cdot v_n}$$

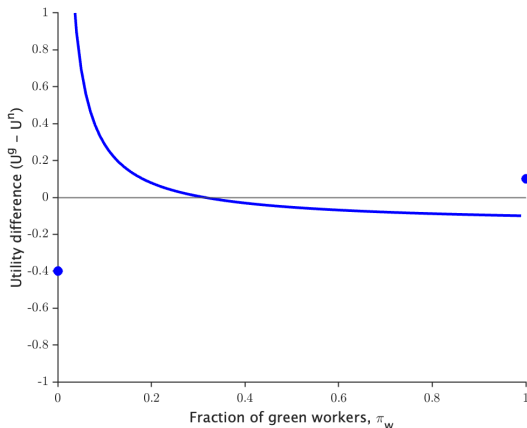


Figure 1: Multiplicity of equilibria in the static benchmark.

Notes: The blue curve shows the utility advantage of green work relative to non-green work as the share of green workers increases. The horizontal zero line marks indifference; crossings are equilibria. Dots indicate a low-green equilibrium, an interior equilibrium, and a high-green equilibrium.

One Shot CRS Equilibria: Comparative Statics

Individual policy margin can expand green employment

- Lower worker entry cost: $\downarrow \kappa_g \Rightarrow \uparrow L_g$ (equivalently, $\uparrow s_g$ lowers effective κ_g)
- Higher firm production subsidy: $\uparrow \tau_g \Rightarrow \uparrow L_g$

$$\frac{\partial L_g}{\partial \kappa_g} < 0, \quad \frac{\partial L_g}{\partial s_g} > 0, \quad \frac{\partial L_g}{\partial \tau_g} > 0.$$

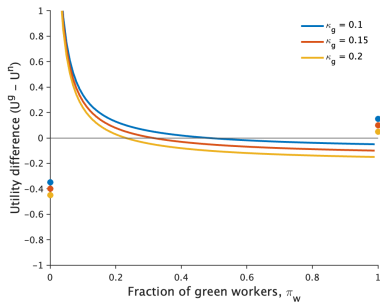


Figure 2: Higher training cost $\Rightarrow$ lower green entry/employment L_g .

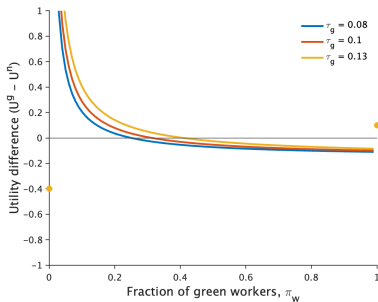


Figure 3: Higher green subsidy $\Rightarrow$ higher green entry/employment L_g .

Dynamic Model: Unemployment, Policies, Welfare

- Discrete time, infinite horizon; discount rate β ; separations at rate λ .
- Labor force is normalized to 1, homogeneous workers
- Workers choose to go either green or non-green³
 - ▶ There is entry cost κ_g in green sector.
- A large number of homogeneous firms decides whether to go green or non-green
 - ▶ green firms receive production subsidy τ_g
 - ▶ measure of active firms determined by free entry

³Labor market segmentation, though might seem an extreme assumption, ? show that in 2021, only 0.7 percent of workers who transitioned out of a dirty job transitioned into a green job.

As per WorkingNation [2024], in the US, green jobs are expected to expand from current 2% to nearly 24 million, comprising 14% of total US jobs by 2030.

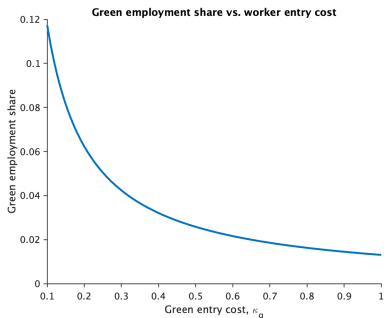


Figure 4: Decreasing worker entry cost increases equilibrium green employment share.

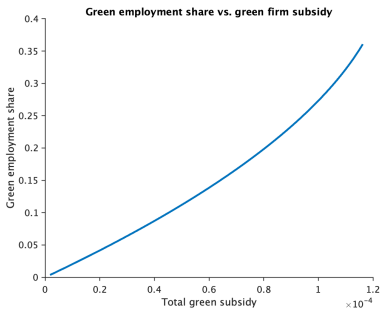


Figure 5: Increasing firm subsidy increases equilibrium green employment share.

All policies achieve the target

How to achieve the target of green employment to move from current 2% to 14% of total US jobs by 2030?

Table 1: Alternative Policy Mixes Achieving 14% Green Employment

Policy configuration	Worker subsidy (% of GDP)	Firm subsidy (% of GDP)	Total subsidy (% of GDP)	Green jobs (%)
Dual intervention	0.48	0.07	0.55	14.0
Worker-only policy	0.60	0.00	0.60	14.0
Firm-only policy	0.00	0.58	0.58	14.0
Baseline (2022)	0.00	0.00	0.00	2.0

⇒ Either lever alone or both together can reach the target.

Two-sided policy dominates across outcomes

Table 2: Policy trade-offs at 14% green employment

Policy configuration	Unemployment (%)	Fiscal cost (% of GDP)	Welfare (CEV) (vs. baseline, %)
Dual intervention	4.77	0.55	-0.84
Worker-only policy	5.16	0.60	-1.11
Firm-only policy	4.80	0.58	-1.07
Baseline (2022)	3.50	0.00	0.970 (level)

- **Unemployment:** Dual vs. one-sided
 - ▶ -39 bps vs. worker-only & -3 bps vs. firm-only
- **Fiscal cost:** Dual vs. one-sided
 - ▶ -5 bps of GDP vs. worker-only & -3 bps vs. firm-only
- **Welfare (CEV):** Dual vs. one-sided
 - ▶ +27 bps vs. worker-only & +23 bps vs. firm-only

⇒ Dual policy achieves the 14% target with *lower unemployment*, *lower fiscal cost*, and *lower welfare distortion* than either one-sided policy.

Green transition is welfare improving only above a threshold

- Without externality:

$$W_0 = p \cdot (e_g + e_n) + (z - \kappa_g)u_g + z \cdot u_n - c \cdot (v_n + v_g)$$

- With positive externality from green employment:

$$W_\psi = p \cdot (e_g + e_n) + (z - \kappa_g)u_g + z \cdot u_n - c \cdot (v_n + v_g) + \phi \cdot e_g$$

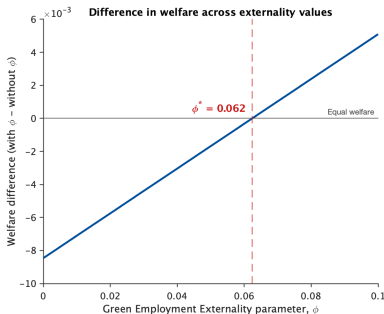


Figure 6: Welfare gains as a function of the externality parameter ϕ .

Aggregate welfare gain/loss depends on ϕ ; welfare equalizes at $\phi^* = 0.062$.

How to interpret the size of the threshold?

Consumption-Equivalent Threshold

- Compare welfare without externality (W_0) to welfare at the threshold (W_{ϕ^*} , $\phi^* \approx 0.062$).
- Compute the constant % increase in consumption that makes $W_0 = W_{\phi^*}$ (CEV).

Result: $\text{CEV} \approx 0.87\%$ at $\phi^* \approx 0.062$.

$\Rightarrow$ Environment benefits need be as big as increasing aggregate consumption by $\approx 0.87\%$ for transition to be aggregate welfare-improving.

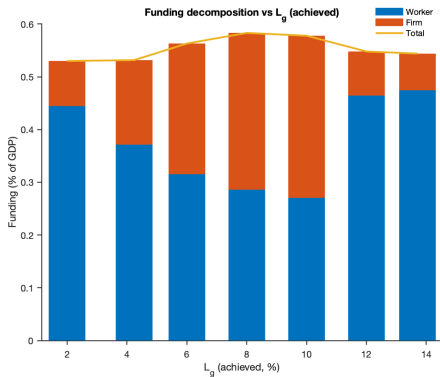


Figure 7: Funding decomposition (worker vs. firm) along the same path; line shows total funding as a percent of GDP.

⇒ Total fiscal cost is hump-shaped over the transition path with rotating policy mix: worker → firm → worker (early → mid → late).

- **Two-sided externality & complementarity**
 - ▶ Workers and firms are strategic complements; each side under-invests.
 - ▶ Using *both* instruments best breaks the coordination friction.
- **State-contingent mix along the path**
 - ▶ **Early:** entry is scarce $\Rightarrow$ tilt to worker support.
 - ▶ **Middle:** market tight; filling constraints bind $\Rightarrow$ tilt to firm support.
 - ▶ **Late:** workers small $\Rightarrow$ tilt back to worker support.
 - ▶ **Rule:** Target the *binding margin*; rotate worker-firm mix as tightness changes.
- **Welfare**
 - ▶ Trade-off: private reallocation costs vs. non-internalized environmental benefits.
 - ▶ Net welfare rises when benefit $>$ cost.

Conclusion & policy

- How to design policy to hit the green labor target efficiently?
- Dual-sided policy beats one-sided alternatives:
 - ▶ Unemployment: -39 to -2 bps
 - ▶ Fiscal cost: -5 to -3 bps of GDP
 - ▶ Welfare (CEV): +23 to +27 bps
- Design rule along the path:
 - ▶ Total support is hump-shaped.
 - ▶ Mix rotates with the binding margin: worker $\rightarrow$ firm $\rightarrow$ worker.
- Welfare threshold: environmental benefit $\geq 0.87\%$ in consumption-equivalent terms $\Rightarrow$ transition raises aggregate welfare.
- Policy recommendation:
 - ▶ Use dual instruments: worker-side entry support and firm-side production support.
 - ▶ Set the mix state-contingently to the binding margin.

Takeaway: A dual, state-contingent policy hits the green labor target with less unemployment, lower fiscal cost, and higher welfare.

Workers shortage amidst green jobs

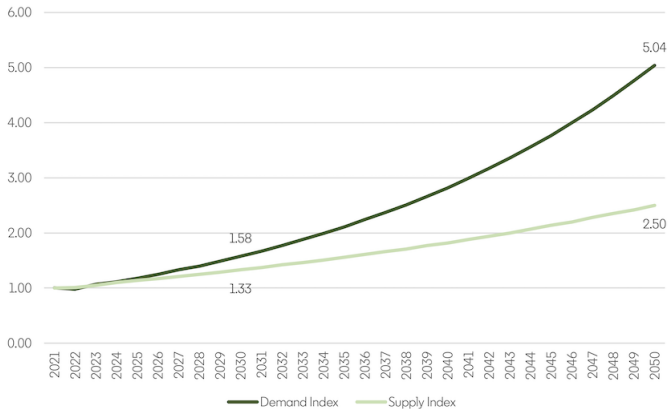


Figure 8: Source: LinkedIn Economic Graph 2024

Model Choice: Entry Costs (κ_g)

What could κ_g be?

Reskilling Needs: Green jobs require new skills, increasing transition costs for workers [??].

Geographic Frictions: Green jobs are regionally mismatched with fossil fuel hubs [??].

Non-Wage Benefits: Lower union protections and fewer benefits in green jobs [?].

Job Uncertainty: Green jobs often involve intermittent or short-term work [?].

Takeaway: Despite wage premium and high matching likelihood in green sector, there is green workers shortage, so κ_g represents the broad costs workers face when shifting to green jobs.

Model Choice: Green Subsidies (τ_g)

IRA Subsidies: The Inflation Reduction Act focuses on firm subsidies but lacks reskilling programs [?]; It provides tax credits for renewable energy, ranging from \$5/MWh to \$32/MWh based on eligibility [Bushnell and Smith, 2024].

Key Issue: Current policy misaligns costs and benefits; A policy-driven transition can help address market frictions [Oei et al., 2020].

One Shot Model ctd.

- *Matching technology*: The matching technology for market j is:

$$f_j(u, v) = \delta_j \left(\frac{uv}{u + v} \right)^{1-\psi} (uv)^\psi, \quad j \in \{n, g\}, \quad \psi \in [0, 1]$$

where u and v are the measures of unemployed workers and vacant jobs.

- For CRS ($\psi = 0$), the matching probabilities are:

$$\alpha_{wg} = \frac{v_g}{\pi + v_g}, \quad \alpha_{wn} = \frac{v_n}{1 - \pi + v_n}$$
$$\alpha_{fg} = \frac{\pi}{\pi + v_g}, \quad \alpha_{fn} = \frac{1 - \pi}{1 - \pi + v_n}$$

One Shot Model Equilibrium

Endogeneous variables: $\pi, v_g, v_n, w_g, w_n, \tau_n$

Equilibrium conditions:

$$c = \frac{\pi}{\pi + v_g} (y + \tau_g - w_g) \quad (1)$$

$$c = \frac{1 - \pi}{1 - \pi + v_n} (y - \tau_n - w_n) \quad (2)$$

$$w_g = \theta(y + \tau_g) + (1 - \theta)z \quad (3)$$

$$w_n = \theta(y - \tau_n) + (1 - \theta)z \quad (4)$$

$$\tau_n = \tau_g \cdot \frac{\frac{\pi}{(\pi + v_g)} \cdot v_g}{\frac{(1 - \pi)}{(1 - \pi + v_n)} \cdot v_n} \quad (5)$$

$$\pi = \begin{cases} 0 & \text{if } G < N, \\ \in (0, 1) & \text{if } G = N, \\ 1 & \text{if } G > N. \end{cases} \quad (6)$$

$$G = -\kappa_g + \frac{v_g}{\pi + v_g} w_g + \left(1 - \frac{v_g}{\pi + v_g}\right) z \quad \text{and} \quad N = \frac{v_n}{1 - \pi + v_n} w_n + \left(1 - \frac{v_n}{1 - \pi + v_n}\right) z$$

Theoretical Model

1. Workers

► Unemployed

$$U = \max\{U_g, U_n\}$$

$$U_g = z - \kappa_g + \beta [\alpha_{wg}W_g + (1 - \alpha_{wg})U_g]$$

$$U_n = z + \beta [\alpha_{wn}W_n + (1 - \alpha_{wn})U_n]$$

2. Employed

$$W_g = w_g + \beta [(1 - \lambda)W_g + \lambda U]$$

$$W_n = w_n + \beta [(1 - \lambda)W_n + \lambda U]$$

3. Firms⁴

► Vacant

$$V = \max\{V_g, V_n\}$$

$$V_g = -c + \beta [\alpha_{fg}J_g + (1 - \alpha_{fg})V]$$

$$V_n = -c + \beta [\alpha_{fn}J_n + (1 - \alpha_{fn})V]$$

► Filled

$$J_g = (1 + \tau_g)p - w_g + \beta [\lambda V + (1 - \lambda)J_g]$$

$$J_n = p - w_n - \tau_n + \beta [(1 - \lambda)J_n + \lambda V]$$

⁴In this framework, allowing firms to choose between green and brown sector is analogous to allowing for free entry in both sector which is what I do hereon.

Solving the model

- Bargaining and wage curves

1. Non-green jobs: $(1 - \eta)(W_n - U_n) = \eta J_n$

$$\implies w_n = \frac{(1 - \eta)z[1 - \beta(1 - \lambda)] + \eta(p - \tau_n)[1 - \beta(1 - \lambda - \alpha_{wn})]}{[1 - \beta(1 - \lambda - \eta\alpha_{wn})]}$$

i.e. $\tau_n \uparrow \implies w_n \downarrow$

2. Green jobs: $(1 - \eta)(W_g - U_g) = \eta J_g$

$$\implies w_g = \frac{(1 - \eta)(z - \kappa_g)[1 - \beta(1 - \lambda)] + \eta p(1 + \tau_g)[1 - \beta(1 - \lambda - \alpha_{wg})]}{[1 - \beta(1 - \lambda - \eta\alpha_{wg})]}$$

i.e. $\tau_g \uparrow \implies w_g \uparrow$ and $\kappa_g \uparrow \implies w_g \downarrow$.

- Optimal Choices for agents

$$U_g = \frac{[1 - \beta(1 - \lambda)][z - \kappa_g] + \beta\alpha_{wg}w_g}{(1 - \beta)(1 - \beta(1 - \alpha_{wg} - \lambda))}$$

$$U_n = \frac{[1 - \beta(1 - \lambda)]z + \beta\alpha_{wn}w_n}{(1 - \beta)(1 - \beta(1 - \alpha_{wn} - \lambda))}$$

$$U_g^* = U_n^*$$

Equilibrium Definitions

A steady state equilibrium comprises of wages (w_g, w_n) , measure of green and traditional vacant firms (v_g, v_n) , measure of green and traditional unemployed and employed workers (u_g, u_n, e_g, e_n) , fraction of unemployed workers who choose to be green (π) , and a green production subsidy τ_g given a flat tax τ_n that satisfy the beveridge curves, wage curves, job creation curves, agents' optimal choices, and government budget constraint.

Equilibrium Equations

1. Beveridge Curves:

- ▶ $\alpha_{wg}u_g = \lambda e_g$
- ▶ $\alpha_{wn}u_n = \lambda e_n$
- ▶ $u_n + u_g + e_n + e_g = 1$

2. Wage Curves

- ▶ $w_n = \frac{(1-\eta)z[1-\beta(1-\lambda)] + \eta(p-\tau_n)[1-\beta(1-\lambda-\alpha_{wn})]}{[1-\beta(1-\lambda-\eta\alpha_{wn})]}$
- ▶ $w_g = \frac{(1-\eta)(z-\kappa_g)[1-\beta(1-\lambda)] + \eta p(1+\tau_g)[1-\beta(1-\lambda-\alpha_{wg})]}{[1-\beta(1-\lambda-\eta\alpha_{wg})]}$

3. Job Creation Curves:

- ▶ $w_g = (1 + \tau_g)p - \frac{c(1-\beta(1-\lambda))}{\beta\alpha_{fg}}$
- ▶ $w_n = p - \tau_n - \frac{c(1-\beta(1-\lambda))}{\beta\alpha_{fn}}$

4. Agents' Optimal Choices:

- ▶ $U_n^* = U_g^*$ i.e. $\frac{[1-\beta(1-\lambda)]z + \beta\alpha_{wn}w_n}{(1-\beta)(1-\beta(1-\alpha_{wn}-\lambda))} = \frac{[1-\beta(1-\lambda)][z-\kappa_g] + \beta\alpha_{wg}w_g}{(1-\beta)(1-\beta(1-\alpha_{wg}-\lambda))}$

5. Government's Budget Constraint

- ▶ $\tau_n = \tau_g \cdot \frac{\alpha_{fg} \cdot v_g}{\alpha_{fn} \cdot v_n}$

Remember:

- $\alpha_{wg} = \delta_g \left(\frac{v_g}{(u_g + v_g)^{1-\psi}} \right)$
- $\alpha_{wn} = \delta_n \left(\frac{v_n}{(u_n + v_n)^{1-\psi}} \right)$
- $\alpha_{fg} = \delta_g \left(\frac{u_g}{(u_g + v_g)^{1-\psi}} \right)$
- $\alpha_{fn} = \delta_n \left(\frac{u_n}{(u_n + v_n)^{1-\psi}} \right)$

Endogenous variables: $\{v_g, v_n, u_g, u_n, e_g, e_n, w_g, w_n, \tau_g\}$

Parameters: $\{\beta, p, \psi, \eta, c, \lambda, z, \delta_g, \delta_n, \kappa, \tau_g\}$

Model Calibration

- Exogeneously set parameters
 - ▶ $\beta = 0.9959$ (discount rate)
 - ▶ $p = 1$ (worker productivity)
 - ▶ $\eta = 0.72$ (worker's bargaining for CRS case)
 - ▶ $z = 40\%$ of average productivity
- Data Targets:
 - ▶ Wage Premium: 2% (IMF, 2022) ⁵.
 - ▶ Green (Renewable Sector) Employment Share: 2% of U.S. workforce (EIA, 2022).
 - ▶ Hiring Likelihood Ratio: Workers with green skills 29% more likely to be hired [LinkedIn Economic Graph, 2023].
 - ▶ Labor Market Tightness: 1.868 (FRED, 2024).
 - ▶ Unemployment Rate: 3.5% (BLS, 2023).
 - ▶ Green Tax Subsidy: 0.01788% - 0.11445% of U.S. GDP (IRA 2022, net renewables).

⁵Estimates up to 4% (CEPR, 2023).

Model Calibration

- Calibration

<i>Data Moments</i>	<i>Model Analogue</i>	<i>Model Values</i>	<i>Target Values</i>
Wage Premium	w_g/w_n	1.02	1.02
Employment Share	$e_g/(e_n + e_g)$	0.0195	0.0195
Hiring Likelihood Ratio	α_{wg}/α_{wn}	1.29	1.29
Labor Market Tightness	$(v_n + v_g)/(u_n + u_g)$	1.868	1.868
Unemployment Rate	$u_n + u_g$	3.5%	3.5%

Table 3: Matching the calibration targets.

- Calibrated parameters

<i>Parameter</i>	<i>Description</i>	<i>Value</i>
c	Vacancy creation cost	0.1223
λ	Separation rate	0.0192
δ_g	Matching efficiency (green)	0.8476
δ_n	Matching efficiency (non-green)	0.8165
κ_g	Worker entry cost (green)	0.7146

Table 4: Internally calibrated parameters.

Transition Policy Design

Goal: Trace policy mix and cost along the path of L_g from 2% $\rightarrow$ 14% .

- **Method:** Study the transition as a *sequence of steady states* at milestones.
- **Why steady states?**
 - ▶ Relevant: Policy is set via dated targets (NDC-style).
 - ▶ Observable: gives point-in-time budgets and labor impacts.
- **Report at each milestone:** Jobs; Policy cost; Outcomes

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